

Point of Care Ultrasound in Nephrology

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DEFINITION AND SCOPE

Point-of-care ultrasonography (POCUS) is a limited ultrasound examination performed by the clinician at the patient's bedside to answer focused questions. Most of these questions are binary in nature; for example, "Does the patient have hydronephrosis?" "Is there a pericardial effusion?" In conjunction with history and physical examination findings, POCUS helps to arrive at a diagnosis or narrow the differential diagnosis. Although it may expedite care and eliminate the need for further imaging, POCUS is not a replacement for comprehensive ultrasound studies. The latter usually involves detailed assessment of an anatomic region and a report that includes a set of standard parameters, whereas POCUS answers focused clinical questions to guide the diagnostic process and management.

The scope of nephrologist-performed POCUS has recently expanded. Kidney ultrasound; volume status assessment using focused cardiac, lung, and abdominal vein Doppler ultrasound; and vascular access ultrasound are all within the scope of contemporary nephrology practice.^{1,2} Fig. 5.1 illustrates the common nephrology-relevant POCUS applications. POCUS has evolved as one of the pillars of bedside clinical examination.^{3,4}

BASIC PHYSICS AND INSTRUMENTATION

The *piezoelectric crystals* in the ultrasound transducer produce high-frequency sound waves and convert the returning echoes (from the organs) into electrical signals, which are then displayed as a two-dimensional grayscale image on the monitor. Although some newer handheld ultrasound devices use chip technology instead of piezoelectric crystals, the wave characteristics, modes, and principles of image acquisition essentially remain the same.

Ultrasound Imaging Modes

Various modes of ultrasound display allow evaluation of different aspects of a tissue. The key modes pertinent to POCUS are described below.

B-Mode

B, or the brightness mode, denotes the regular two-dimensional (2D) grayscale display composed of several dots of varying brightness. It is the universal imaging mode, where structures appear in different shades of gray from black to white (Fig. 5.2A).

M-Mode

M, or the motion mode, is a unidimensional imaging mode that displays the movement of a structure over time. It is used to detect and/or quantify the movement of structures (e.g., respiratory variations in

the diameter of the inferior vena cava [Fig. 5.2B], excursion of cardiac valves, pleural sliding).

Color Doppler

In color flow mode, the velocity of mobile acoustic interfaces (red blood cells [RBCs]) is measured as a shift in frequency and represented as a range of colors. This mode is used to detect the presence of flow and identify its direction. When the RBCs move away from the transducer, the reflected frequency is lower than the transmitted frequency, and the difference or shift is represented as *blue*. On the other hand, if the RBCs move toward the transducer, reflected frequency is higher than the transmitted frequency and the shift is represented as *red*. The detection of frequency shift is optimal when the angle of insonation (direction of the ultrasound beam) is parallel to the flow, whereas a perpendicular angle (90 degrees) results in minimal color display with ambiguous directionality. Fig. 5.2C illustrates the arterial and venous flow in a renal allograft using color Doppler.

Spectral Doppler

This mode enables quantitative assessment of the blood flow by graphically representing velocity over time. The spectral waveform is displayed above the baseline if the flow is toward the transducer (analogous to red in color mode) and below the baseline if the flow is away from the transducer (blue in color mode). Spectral Doppler is of two main types: pulsed wave (PW) and continuous wave (CW) Doppler. PW Doppler involves emission of ultrasound waves in pulses and is used to quantify blood flow at a particular location (e.g., to measure the velocity in the renal artery, to assess the flow pattern in portal vein). Figure 5.2D illustrates PW Doppler tracing of a renal artery.

CW Doppler involves continuous emission of ultrasound waves and can measure higher velocities than PW Doppler. However, it does not sample a specific area and displays the signals received from all the vessels along the path of the Doppler line. This mode is useful in echocardiography when accurate measurement of peak velocity across a region is needed, such as the assessment of valvular stenosis.

Transducer Selection

The transducer or probe is a component of the ultrasound machine that contains piezoelectric crystals and is used to acquire images. There are various types of transducers that differ by frequency, crystal arrangement, and shape. High-frequency waves produce high-resolution images but penetrate shallower depths. Conversely, low-frequency waves penetrate deeper at the expense of resolution. Therefore high-frequency transducers are typically used to image superficial structures such as an arteriovenous fistula, and low-frequency transducers are used for deeper structures such as the kidney and the heart. For

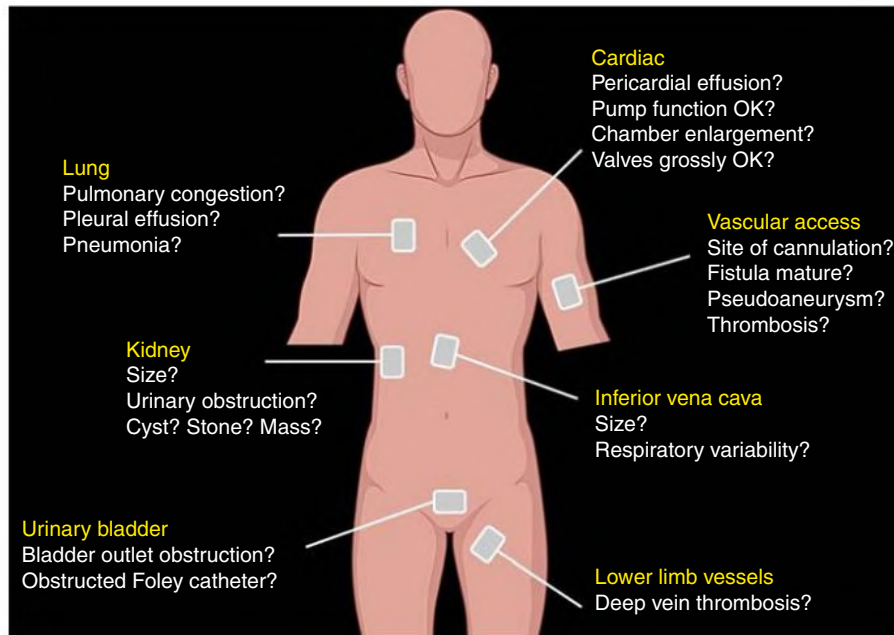


Fig. 5.1 Common sonographic applications and focused questions encountered in nephrology practice. (Created with [Biorender.com](https://www.biorender.com/).)

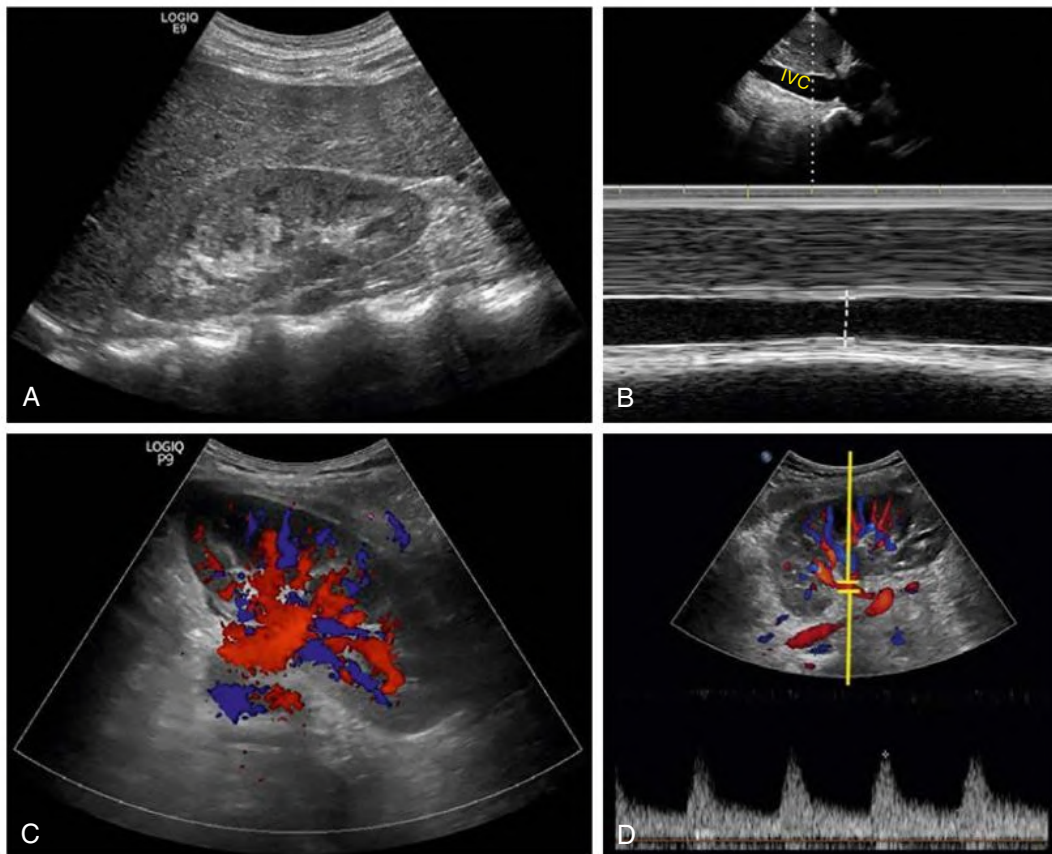


Fig. 5.2 (A) Grayscale image of a normal kidney. (B) M-mode image of the inferior vena cava. Tracing represents movement of the structure over time along the cursor (*dotted line*). (C) Color Doppler image of a normal transplanted kidney. (D) Pulsed wave Doppler image of renal artery. Note the Doppler line, which is like the M-mode cursor but with a small opening called sample volume, which should be placed in the region of interest to measure the flow. (Created with [Biorender.com](https://www.biorender.com/).)

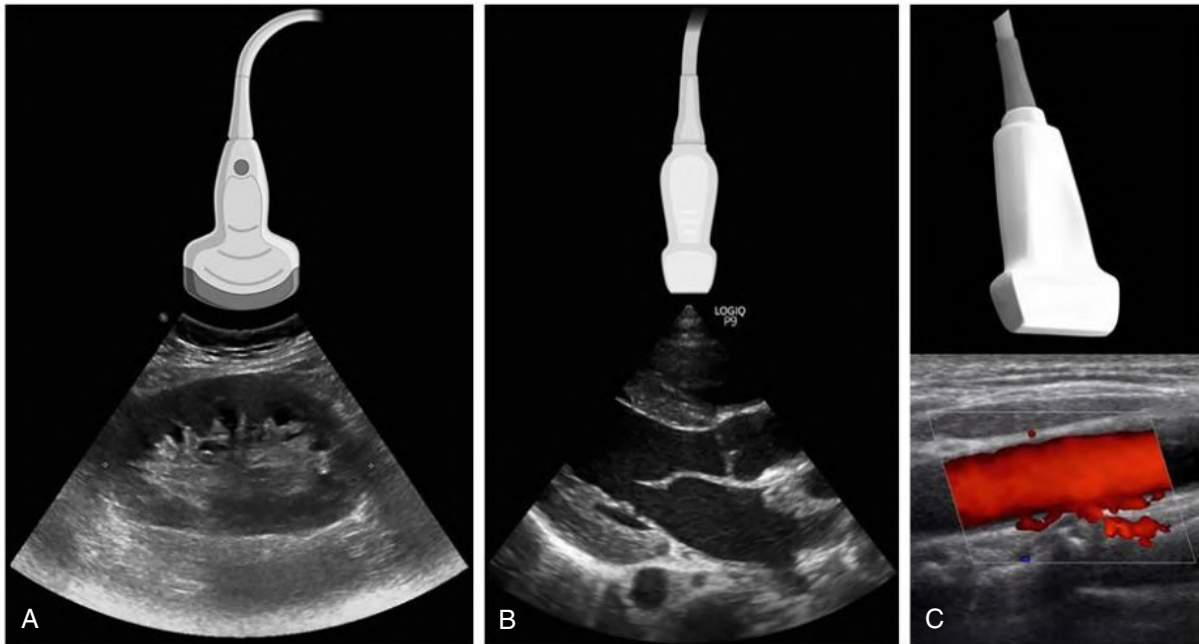


Fig. 5.3 Curvilinear (A), phased array (B), and linear (C) transducers illustrating their respective fields of view. (Created with [Biorender.com](https://www.biorender.com/).)

nephrology-related applications, the three most common types of transducers are curvilinear, phased array, and linear. A *curvilinear transducer* is a low-frequency probe that consists of curvilinear or convex arrangement of crystals and produces a broad field of view (Fig. 5.3A). This allows scanning wider areas and deeper structures; hence it is best suited for abdominal applications. A *phased array transducer* is also a low-frequency transducer but with a small footprint and central arrangement of crystals producing a pie-shaped image (Fig. 5.3B). Because this enables scanning through narrow spaces such as rib interspaces, it is well suited for cardiac applications. Moreover, *phased*, or differential, excitation of piezoelectric elements produces sequential pulses of ultrasound waves, which allows optimal imaging of mobile structures (e.g., heart). A *linear transducer* is a high-frequency probe with crystals arranged in a straight line and produces a rectangular image (Fig. 5.3C). It is best suited for vascular applications, which require higher resolution. Some newer handheld ultrasound devices combine these different properties into one transducer, allowing multisystem scanning by changing the presets (e.g., abdomen, vascular, cardiac) without actually switching the probe.

Image Orientation and Interpretation

Understanding the image orientation is key to accurate interpretation. A structure that is closer to the transducer is displayed at the top of the image, whereas a structure that is farther from the transducer is displayed at the bottom. What is displayed on the right and left of the image depends on the direction of transducer orientation marker (probe marker) while obtaining the image. This marker can be a dot, a light, or a ridge on the transducer and corresponds to the orientation marker on the screen (screen indicator), which is toward the left in the abdomen preset. During a standard abdominal exam, the probe marker is pointed toward the patient's head when obtaining longitudinal images. Therefore the superior pole of the kidney is on the left and the lower pole is on the right of the image in the long axis view. Fig. 5.4 illustrates image orientation.

The displayed echogenicity or brightness of a tissue or structure depends on the strength of the reflected signals. Structures that efficiently transmit sound waves without reflection appear black and

are described as *anechoic* (i.e., not making echoes). Examples include urine, blood, bile, and serous fluid in cysts. On the other hand, *hyperechoic* structures reflect more sound waves (make more echoes) and appear bright/white. Examples include calcified and fibrous structures such as stones, diaphragm, and pericardium. Some hyperechoic structures (e.g., stones, bones) create shadows due to near-total reflection of ultrasound waves precluding visualization of the underlying structures. Air is also a poor ultrasound transmitter, and its presence will preclude imaging of deeper structures. Most soft tissues appear in different shades of gray and are described in terms of *relative echogenicity*, that is, in comparison with the brightness of surrounding structures. Structures that reflect fewer sound waves compared to surrounding structures are described as being *hypoechoic* (e.g., renal cortex vs. liver). The term *isoechoic* is used to describe structures that make echoes of similar intensity to that of the surrounding structures. For example, cortex of the kidney can be isoechoic to that of the medulla in some cases of chronic kidney disease, where corticomedullary differentiation is lost.

KIDNEY AND BLADDER ULTRASOUND

Technique

Kidney ultrasound is performed using a curvilinear transducer. If the latter is not available, a phased array transducer can be used to rule out gross abnormalities such as hydronephrosis; however, parenchymal characteristics and small lesions may not be well visualized. Liver and spleen act as *acoustic windows* to image the right and left kidneys, respectively—meaning that the right kidney is visualized through the liver and the left kidney is visualized through the spleen; otherwise, bowel gas will impede the view. Examination is performed with the patient in the supine position, though lateral decubitus, oblique, or even prone positions may sometimes be required for visualization of the left kidney due to overlying bowel gas. To acquire longitudinal views of the kidney, the transducer is positioned in the mid or anterior axillary line at the tenth rib interspace, with the orientation marker pointing toward the patient's head and aiming slightly posterior. The transducer may have to be moved superiorly or inferiorly depending

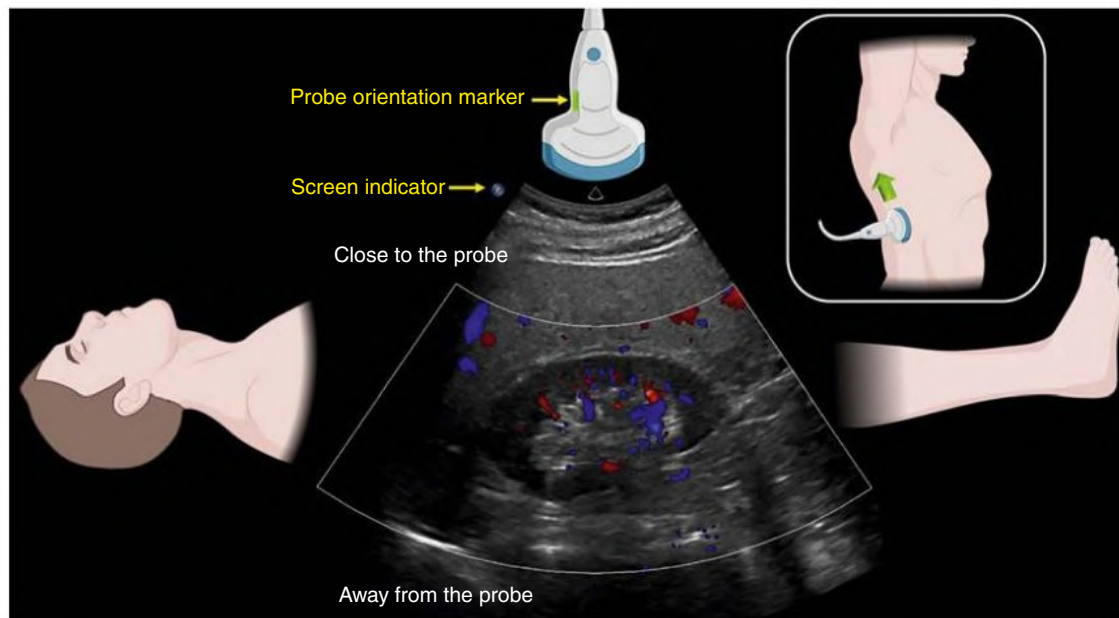


Fig. 5.4 Orientation to an Ultrasound Image. The *green arrow* denotes direction of the probe orientation marker. The right kidney is shown here, but the principles remain the same for any organ being imaged. (Created with [Biorender.com](https://www.biorender.com).)

on the patient's body habitus. Once the entire long axis of the kidney is in view, the transducer must be fanned (tilted) anteriorly and posteriorly to image it completely. The transducer is then rotated 90 degrees counterclockwise to obtain a short axis or transverse view of the kidney. With the transverse view obtained, the transducer is fanned superiorly and inferiorly to visualize the kidney from pole to pole. Although comprehensive Doppler evaluation to detect renal artery stenosis is beyond the scope of POCUS, color Doppler images of the kidney should be obtained. They provide a global assessment of kidney perfusion and allow detection of gross vascular abnormalities such as an arteriovenous malformation that can sometimes mimic a dilated collecting system.⁵

Image acquisition of the transplanted kidney is relatively easy because of its superficial location in the pelvis. Longitudinal and transverse views should be obtained, paying attention to the surgical anatomy and perinephric space to evaluate for collections such as hematoma or lymphocele. In children and thin individuals, a linear transducer may be used to obtain images with higher resolution.

The urinary bladder is also imaged using a curvilinear transducer with the patient supine. Usually, the transverse view is obtained first by placing the transducer just above the pubic symphysis with the orientation marker to the patient's right and aiming the beam inferiorly. A fanning movement should be performed to visualize the bladder superoinferiorly. The transducer is then rotated 90 degrees clockwise with the orientation marker facing the patient's head to obtain the longitudinal view and fanned to visualize the right and left walls of the bladder.

Interpretation

On sonographic images, the kidney appears as a well-defined, bean-shaped structure surrounded by an echogenic capsule. The kidney is about 9 to 12 cm long in the longitudinal plane and varies with patient height. Kidney parenchyma is composed of the outer cortex and inner medulla, which in turn is organized into multiple pyramids. Cortex is usually *hypoechoic* (or *isoechoic*) compared to the adjacent normal liver and spleen. Cortical tissue normally extends into the medulla, separating pyramids in the form of columns called columns

of Bertin. These columns can sometimes hypertrophy and mimic a neoplasm. The medullary pyramids appear *anechoic* or *hypoechoic* compared to the cortex. These are prominent in young children but not always well visualized in adults. The echogenic sinus fat, which encases the renal collecting system, occupies the majority of the inner kidney. The undilated collecting system is not typically visible as a distinct structure; therefore the appearance of an *anechoic* area amid sinus fat should raise the suspicion for hydronephrosis. On the other hand, the collecting system of a well-functioning transplanted kidney is often slightly dilated, likely because of a combination of an increased volume of urine produced (solitary functioning kidney) and loss of the ureter's tonicity from denervation.⁶ In the transverse or short axis view, the midportion of the kidney is C-shaped, with the vessels entering and leaving through the hilum, and the poles appear circular (Fig. 5.5).

The urinary bladder appears as an *anechoic*, fluid-filled structure (urine is black on ultrasound). It is ovoid to rectangular when full, but the shape varies with the amount of urine present. Note should be made of the uterus in females and the prostate in males where applicable.

When evaluating a patient with acute kidney injury (AKI), excluding urinary obstruction is the main indication for POCUS. In addition, evaluating kidney size and echogenicity gives information about chronicity of the kidney dysfunction (see [Chapter 97](#)).

Hydronephrosis

Hydronephrosis is essentially *anechoic* backlogged urine that appears as branching, interconnected areas in the renal collecting system. The severity of hydronephrosis is qualitatively graded as mild, moderate, or severe (Fig. 5.6). In mild hydronephrosis, there is dilation of the renal pelvis and calyces, but the overall parenchymal architecture is retained. In moderate hydronephrosis, the medullary pyramids start to flatten due to back pressure in addition to increasing dilation of the pelvicalyceal system. In severe cases, the renal pelvis and calyces appear ballooned (occupying most of the kidney), and corticomedullary differentiation is lost (making the parenchyma thin). If the hydronephrotic area demonstrates internal echoes, pyonephrosis (pus in

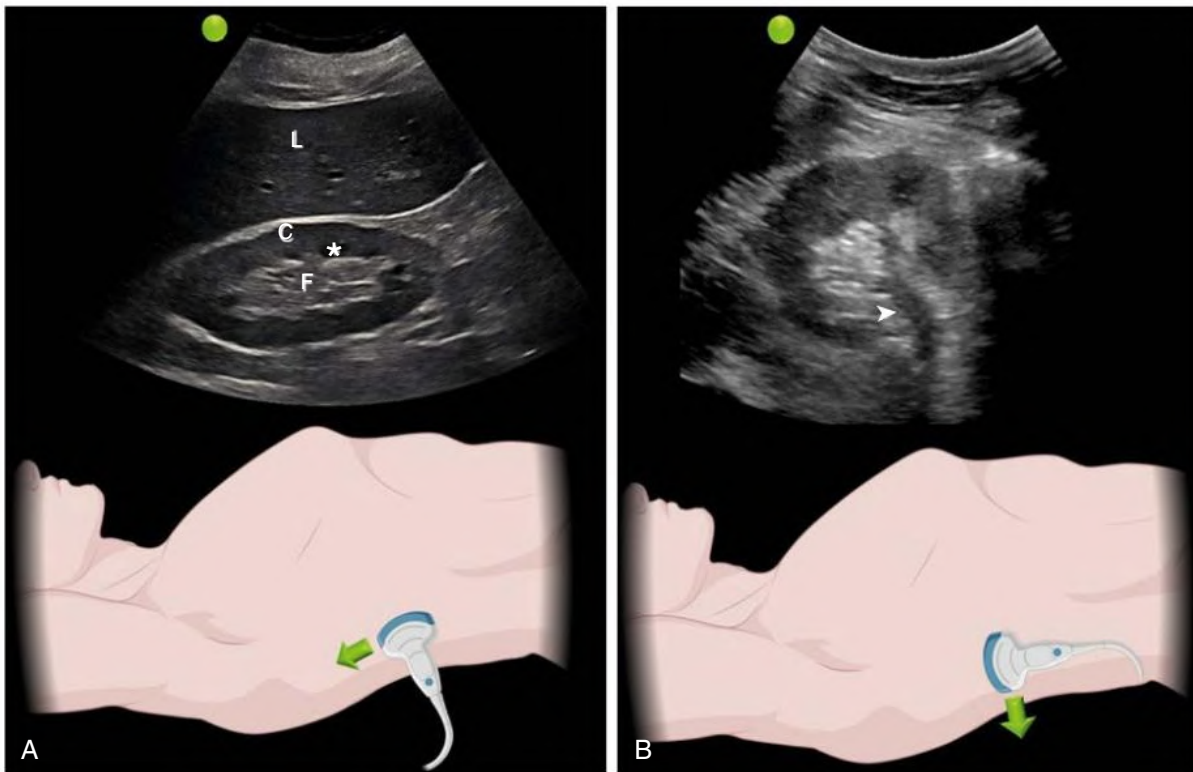


Fig. 5.5 Longitudinal (A) and Transverse (B) Views of a Normal Kidney. The *green arrow* denotes direction of the probe orientation marker during image acquisition. The *asterisk* denotes the medullary pyramid. In the transverse image, the *arrowhead* indicates renal vein (use color Doppler to differentiate from a hydro-ureter). C, Cortex; F, sinus fat; L, liver. (Created with [Biorender.com](https://www.biorender.com).)



Fig. 5.6 Mild (A), moderate (B), and severe (C) hydronephrosis.

the collecting system) should be considered in the appropriate clinical context, such as when the patient presents with fever and flank pain. False negatives for hydronephrosis can occur in the setting of acute or partial obstruction, volume depletion, and retroperitoneal fibrosis.^{7,8} It is prudent to perform repeat POCUS examination and/or excretory urography when the suspicion for obstruction is high.

Prominent renal vasculature and vascular malformations can mimic hydronephrosis, as both blood and urine appear *anechoic*. However, demonstrating the presence of flow using color Doppler confirms vasculature (Fig. 5.7A–B). Similarly, cysts are also *anechoic* and may be confused with hydronephrosis, particularly when close to the renal pelvis (parapelvic cysts) (Fig. 5.7C–D). However, cysts tend to be round and well circumscribed as opposed to hydronephrosis, which is an irregular branching structure. In addition, continuity can be demonstrated between dilated renal pelvis and hydroureter while

the cyst is not connected to the ureter. Comparison with prior imaging helps when available.

Stones

Kidney stones may be encountered in patients with urinary obstruction. On grayscale images, they appear as *hyperechoic* structures with a posterior acoustic shadow. Acoustic shadowing is the black area or signal void seen beyond structures that do not transmit ultrasound waves. In the color Doppler mode, stones exhibit the “twinkling sign,” which refers to a rapidly alternating focus of color Doppler signals mimicking turbulent flow, which is more pronounced with rougher stones.

Urinary Bladder

Sonographic evaluation of urinary obstruction should always include images of the urinary bladder. It helps in assessing urinary retention,

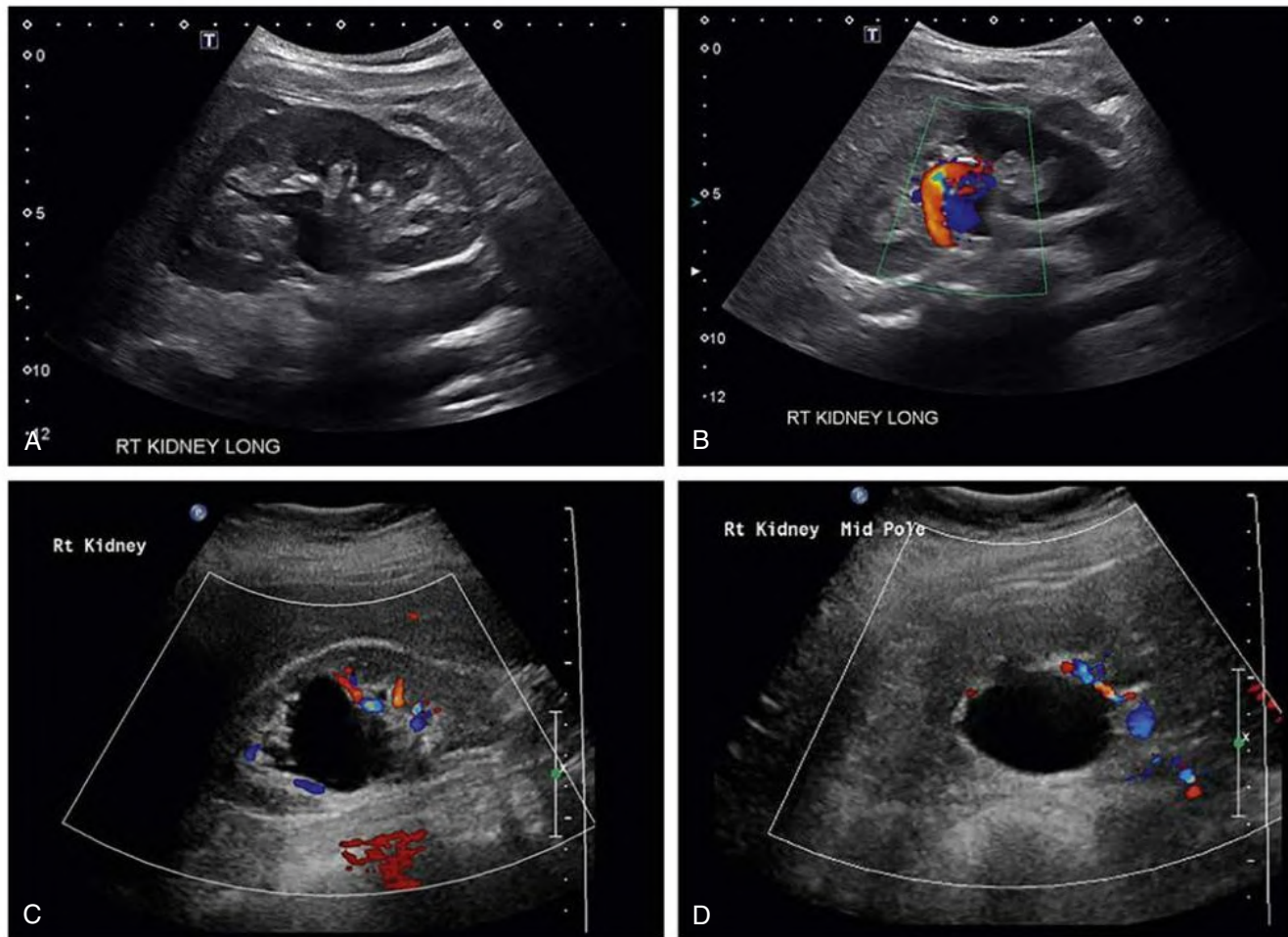


Fig. 5.7 (A) Grayscale image of the kidney demonstrating an anechoic area mimicking mild hydronephrosis. (B) Color Doppler image of the same kidney demonstrating blood flow suggestive of vasculature (arteriovenous malformation in this case). (C–D) Longitudinal and transverse images of kidney demonstrating a large parapelvic cyst mimicking hydronephrosis.

and pathologies such as enlarged prostate, bladder stones, and masses may be found that guide management. Furthermore, a misplaced or obstructed Foley catheter can be easily identified (Fig. 5.8).

FOCUSED CARDIAC ULTRASOUND

Accurate assessment of volume status allows effective management of patients with complex fluid and electrolyte disorders commonly encountered in nephrology practice. Goal-directed ultrasonography of the pump, pipes, and leaks provides helpful insights into a patient's hemodynamics. Pump represents focused cardiac ultrasound (FoCUS), pipes represent inferior vena cava (IVC) ultrasound and venous Doppler, and the leaks indicate assessment of the extravascular lung water and ascites. Though separately mentioned here, IVC ultrasound is often considered a component of FoCUS.

Technique

There are five standard views of FoCUS, namely, parasternal long axis, parasternal short axis, apical four-chamber, subcostal four-chamber, and subcostal IVC. It is important to be familiar with all these views, as some of them may not be well visualized depending on the patient's body habitus, surgical dressings, and so forth, necessitating an alternative approach. The following is a brief description of the image acquisition technique as well as key cardiac structures visualized in each of these

views. Also, note that by convention the screen indicator is displayed to the right in the cardiac preset (as opposed to left in the abdominal and vascular presets) but still corresponds to the probe marker. Therefore failure to select the appropriate preset results in flipped images.

Parasternal Long Axis (PLAX) View

This view images the heart in *its* long axis, which is oblique with respect to the *body's* long axis. The transducer is placed just to the left of the sternum (*parasternal*) in the third or fourth intercostal space with the probe marker toward patient's right shoulder.

In the PLAX image (Fig. 5.9A), the right ventricular outflow tract (RVOT) is seen on the top of the screen as it is closer to the transducer. The long axis view of the left ventricle (LV) is seen posterior to it with its inflow (i.e., the left atrium [LA] and mitral valve) and the outflow (i.e., the aortic valve and ascending aorta [Ao]). Slight rotation and tilting of the transducer may be needed to fully open the LV cavity. The depth of the image should be appropriately adjusted to visualize the descending aorta, which appears as an anechoic circular structure posterior to the LA/LV. This is an important landmark to differentiate pericardial from pleural effusion in this view (see below).

Parasternal Short Axis (PSAX) View

This view images the heart in its transverse plane. It is obtained by rotating the transducer 90 degrees clockwise from the PLAX position

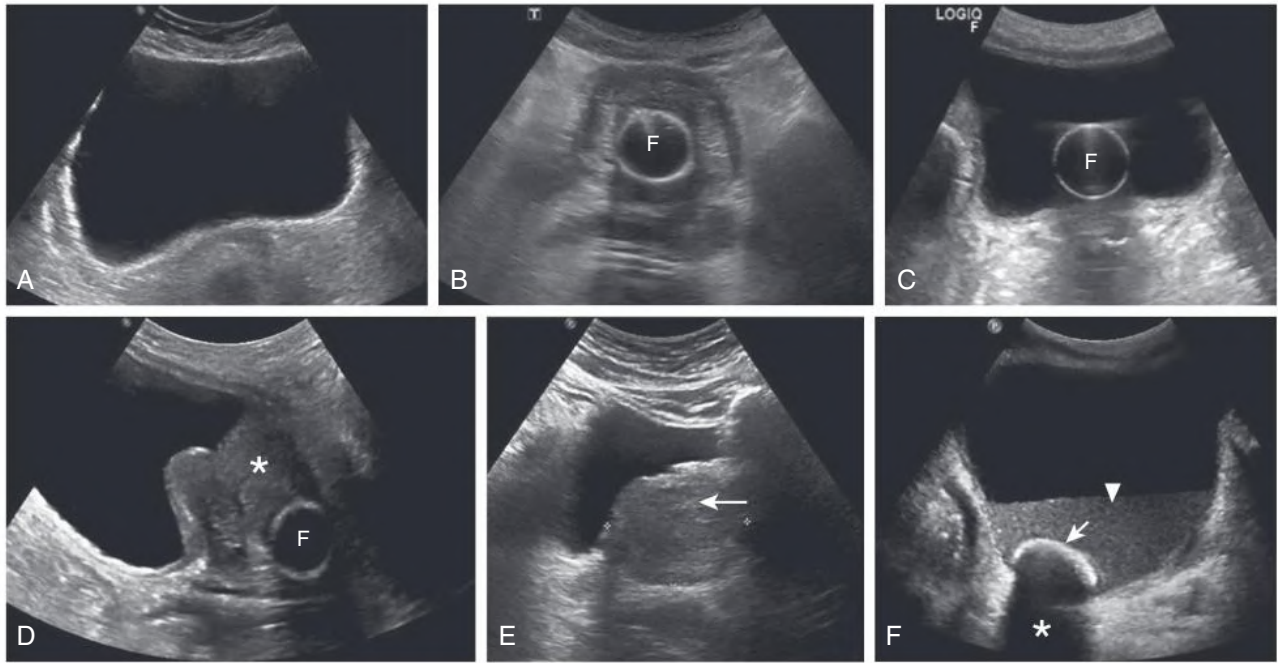


Fig. 5.8 (A) Normal urinary bladder ultrasound, transverse plane. (B) Decompressed bladder with Foley catheter. (C) Bladder filled with anechoic urine despite having a Foley catheter indicative of obstructed catheter. (D) Misplaced Foley catheter balloon in prostatic urethra; *asterisk* indicates prostate gland. (E) Enlarged prostate gland (*arrow*) compressing the bladder. (F) Bladder stone (*arrow*). Note the acoustic shadowing (*asterisk*) and surrounding debris (*arrowhead*). *F*, Fluid-filled balloon of the catheter.

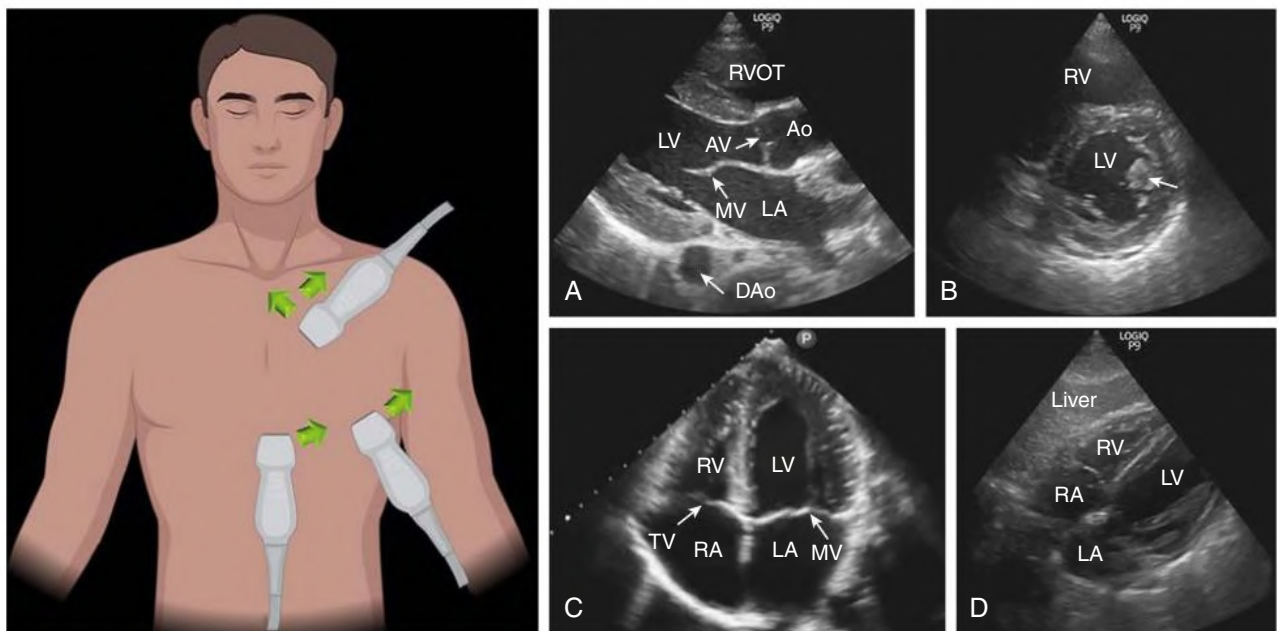


Fig. 5.9 Standard Views of Focused Cardiac Ultrasound. *Left*, Transducer position to acquire these views. *Green arrows* denote direction of the transducer orientation marker. *Right*, Sonographic images of parasternal long axis (A), parasternal short axis (B; *arrow* indicates papillary muscle), and apical (C) and subcostal (D) four-chamber views. *Ao*, Aorta; *AV*, aortic valve; *DAo*, descending thoracic aorta; *LA*, left atrium; *LV*, left ventricle; *MV*, mitral valve; *RA*, right atrium; *RV*, right ventricle; *RVOT*, right ventricular outflow tract; *TV*, tricuspid valve. (Created with [Biorender.com](https://www.biorender.com).)

such that the orientation marker is facing the patient's left shoulder. There are three main levels of visualization in PSAX that can be obtained by tilting the probe superiorly and inferiorly. The aortic valve can be seen by tilting up the transducer toward the patient's right shoulder. The mitral valve level is obtained when the probe is

almost perpendicular to the chest wall. The papillary muscle level comes into view when the face of the probe is slightly tilted down toward the patient's left flank.

Of these, the papillary muscle or the midventricular view is considered a standard FoCUS view and consists of the semilunar right

ventricle (RV) on the top followed by the round LV with papillary muscles appearing as inward projections (Fig. 5.9B). This view allows evaluation of the radial function of the LV and detection of the RV enlargement.

Apical Four-Chamber View

This view images the heart in its coronal plane from the apex. It is obtained by placing the transducer in the fourth or fifth intercostal space (where the apical impulse is felt) with the orientation marker facing the patient's left shoulder (somewhere between 2 o'clock and 3 o'clock positions). Another approach is to slide the transducer infero-laterally from the PSAX position toward the apex. Whenever possible, positioning the patient in the left lateral position improves the image quality; this also applies to parasternal views.

The apical four-chamber view demonstrates all four chambers of the heart and the tricuspid and mitral valves. The right atrium (RA) and RV are on the left of the screen and the LA/LV on the right (Fig. 5.9C). This is a good view to assess atrioventricular valve regurgitation using color Doppler and vegetations, if they are large enough to be seen on transthoracic echo. The apical five-chamber view is obtained by tilting the transducer slightly anterior from the four-chamber view, which brings the LV outflow tract and the aortic valve into view (fifth chamber).

Subcostal Four-Chamber View

This view images the heart through the liver from the subxiphoid area. The transducer is placed just below the xiphoid process in the midline with the orientation marker toward the patient's left side. The transducer should be pressed firmly and tilted such that it is almost parallel to the skin surface. Having the patient bend their knees when possible helps to relax the abdominal wall musculature. This view is particularly useful in patients with difficult or inaccessible chest windows, such as those with hyperinflated lungs (obstructive airway disease, mechanical ventilation), surgical dressings, and so forth.

The subcostal four-chamber view shows the same structures as an apical four-chamber view but appears tilted because a different scan plane is used (Fig. 5.9D). RV size can be underestimated in this view; hence caution should be exercised when commenting on the relative chamber size. In general, it is a good practice to corroborate abnormal findings in at least two different cardiac views.

Subcostal Inferior Vena Cava View

This view is obtained by rotating the transducer approximately 90 degrees counterclockwise from the subcostal four-chamber view. The IVC is seen as a longitudinal anechoic structure joining the heart (RA) (Fig. 5.2B). The hepatic vein is often seen entering the IVC close to the IVC-RA junction. Care must be taken to visualize the widest diameter of the IVC by tilting the transducer from left to right (the ultrasound beam should pass through the center of the vessel). When assessing the RA pressure, maximal diameter and respiratory variation of the IVC are measured just distal to the hepatic vein-IVC confluence, or approximately 2 cm from the IVC-RA junction. IVC size should not be measured at the IVC-RA junction because the diaphragmatic pull can mimic collapse and lead to misinterpretation. A transverse view of the IVC can be obtained by rotating the transducer 90 degrees counterclockwise from this position; this helps to clarify the measurements when in doubt. Care must be taken to avoid mistaking the aorta for the IVC. Contrary to the IVC, the aorta is not in direct contact with the liver, and no IVC-RA junction will be seen.

Interpretation

FoCUS quickly provides the answers to the five Es, that is, the qualitative estimation of left ventricular *ejection*; the presence or absence of

pericardial *effusion*; *equality* (relative ventricular size); *entrance* (IVC size and collapsibility); and the *exit* (stroke volume measurement at the left ventricular outflow tract [LVOT]).^{9,10} In addition, attention should be paid to any gross abnormalities such as intracardiac thrombi and valvular vegetations. Also, a quick color Doppler assessment of atrioventricular valves in the apical four-chamber view is helpful to exclude significant regurgitant lesions leading to pulmonary or systemic vascular congestion.

Ejection

Assessment of LV systolic function, or the *ejection*, is a key component of volume status assessment. Qualitative estimation of or "eyeballing" the ejection fraction is done by observing wall motion and comparison of the chamber size between systole and diastole. Wall motion constitutes both thickening and inward motion of the endocardium during systole. In general, the walls should approximate by one-third or more. In patients with depressed LV function, both wall thickening and inward motion are decreased. On the other hand, a hyperdynamic ventricle, that is, where ventricular walls/papillary muscles in the parasternal short axis view almost touch at end systole, may indicate volume depletion if the overall clinical picture is suggestive. In addition, the movement of the mitral valve should be observed. Normally, the anterior leaflet almost touches the interventricular septum with each heartbeat. If the ejection fraction is low, reduced stroke volume results in decreased blood flow across the mitral valve, and it does not open maximally, increasing the distance between the septum and anterior leaflet (Video 5.1). When visual estimation is ambiguous or the physician does not have enough experience in eyeballing, M-mode measurement of this motion can be helpful. The distance between the peak of mitral valve tracing during early passive diastolic filling (E-point) and interventricular septum of greater than 7 mm is generally considered as a marker of significant LV systolic dysfunction (LVEF <30%).¹¹ This is called E-point septal separation (EPSS).

Effusion

Pericardial *effusion* is an important cause of hypotension and hemodynamic compromise. On ultrasound, it appears as an *anechoic* space between the two pericardial layers. PLAX and subcostal views allow better visualization of the effusion, though it is generally identifiable on all the standard FoCUS views (Fig. 5.10A–B). In the PLAX view, a left pleural effusion may be mistaken for pericardial effusion because the pleural space is immediately posterior to the pericardium. The descending thoracic aorta is an anatomic landmark that helps to differentiate pericardial and pleural effusions in this view. Pericardial effusion is seen anterior to the descending aorta, whereas the pleural effusion is seen posteriorly or starts at the same level (Fig. 5.10C–D). Moreover, a collapsed lung is often seen as a mobile echogenic structure within the pleural effusion.

With respect to severity, a separation between the pericardial layers in diastole of less than 1 cm is considered mild effusion, whereas 1 to 2 cm is considered moderate and greater than 2 cm severe.¹² Whenever a pericardial effusion is detected, note should be made of right atrial collapse at the onset of systole, or diastolic right ventricular collapse. These suggest cardiac tamponade, as does a dilated IVC with minimal or absent respiratory variation.

Equality

Equality refers to relative chamber size, particularly of the RV compared to LV. A normal RV is less than two-thirds the size of the LV and can quickly dilate with pressure or volume overload. When the RV is mildly enlarged, it is slightly more than two-thirds the size of the LV and is of the same size as the LV when moderately enlarged.

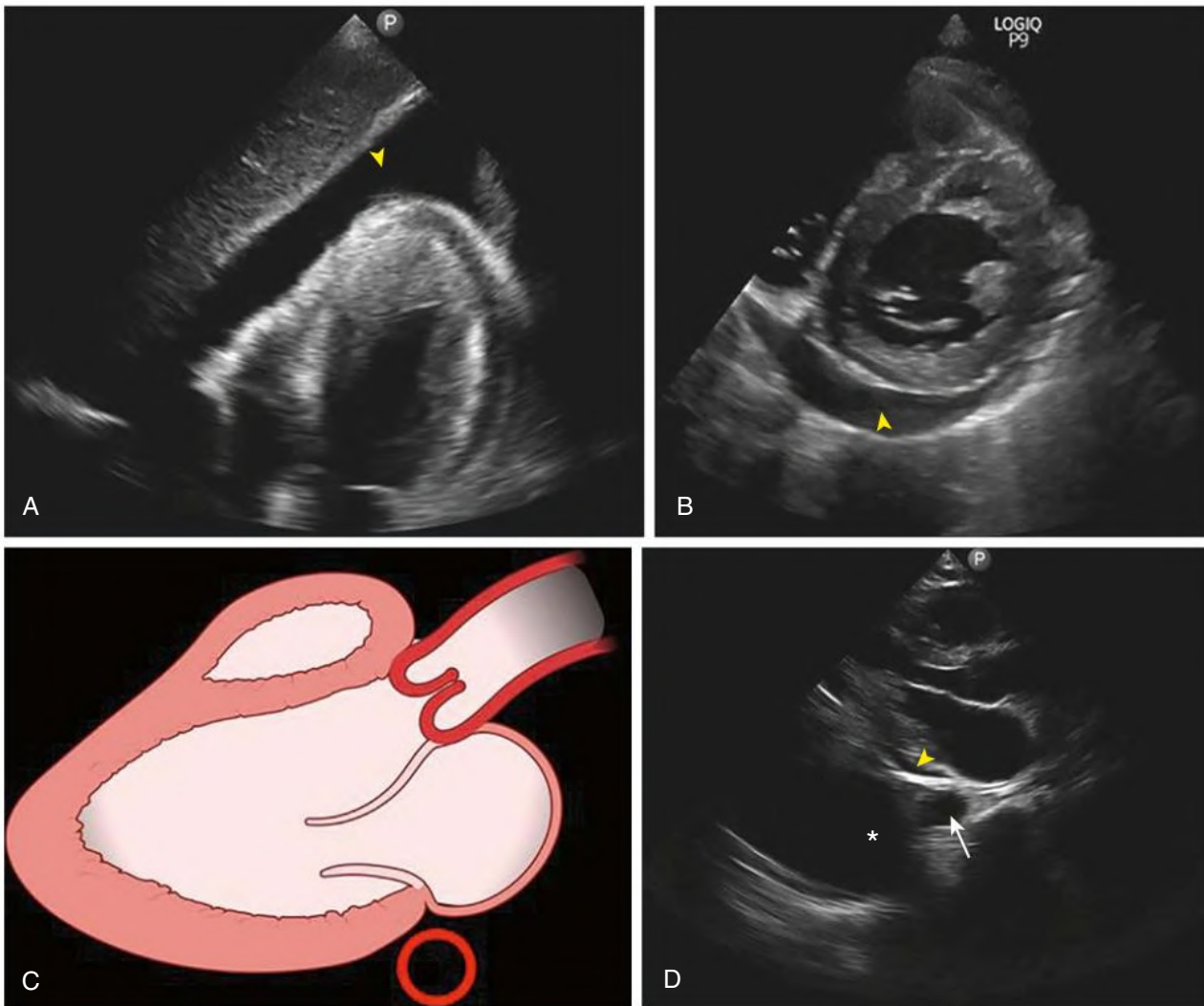


Fig. 5.10 Subcostal (A) and parasternal short axis (B) views demonstrating pericardial effusion (arrowhead). (C) Schematic of normal parasternal long axis view. (D) Parasternal long axis view demonstrating a small pericardial effusion (arrowhead) anterior to the descending aorta (arrow) and a left pleural effusion (asterisk) posterior to it. (C, Created with [Biorender.com](https://www.biorender.com).)

Severe RV dilation is characterized by it being bigger than the LV (Fig. 5.11A). The apical four-chamber and PSAX are good views to assess this. With volume overload, the RV becomes dilated and the interventricular septum is flattened predominantly in diastole, giving a D-shaped appearance to the LV in the PSAX view. This is called the *D-sign* (Fig. 5.11B). This occurs in both systole and diastole if there is pressure overload (e.g., pulmonary embolism or severe pulmonary hypertension). Although these patients are often hypotensive, indiscriminate administration of intravenous fluids results in further compromise of the LV cavity from excess preload-induced RV dilation leading to drop in cardiac output.

Entrance

Entrance (to the heart) refers to evaluation of IVC size and collapsibility to estimate RA pressure, which is a driver of systemic venous congestion and congestive nephropathy.¹³ In spontaneously breathing patients, the IVC collapses during inspiration due to negative intrathoracic pressure. An IVC diameter of 2.1 cm or less and collapsibility greater than 50% with a sudden inspiration (a sniff) indicates normal RA pressure of 3 mm Hg (0–5 mm Hg); an IVC diameter greater than 2.1 cm with less than 50% inspiratory collapse indicates high RA pressure

of 15 mm Hg (10–20 mm Hg); scenarios in between are ascribed to an intermediate value of 8 mm Hg (5–10 mm Hg). These cutoffs cannot be used in mechanically ventilated patients because the IVC is dilated at baseline due to positive pressure ventilation and may not collapse at all during the respiratory cycle. Moreover, not all patients are able to sniff on request and IVC collapsibility can significantly vary with magnitude of the inspiratory effort. In general, IVC ultrasound can be used as an indicator of *fluid tolerance*, that is, as a stop point for administering fluids when the IVC is plethoric. However, assessment of *fluid responsiveness* (increase in cardiac output in response to fluids) should not be based on IVC alone. Bedside stroke volume should be measured when possible as described below. Video 5.2 illustrates a small collapsible IVC and a plethoric IVC.

Exit

Exit (from the heart) refers to measurement of the stroke volume at the LVOT. In the emergency medicine literature, *exit* was originally used to denote assessment of the aortic root for thoracic aortic aneurysm and thoracic aortic dissection.¹⁰ In patients with tenuous fluid status but no gross volume overload, assessing fluid responsiveness using FoCUS is helpful when intravenous fluid therapy is contemplated. The passive

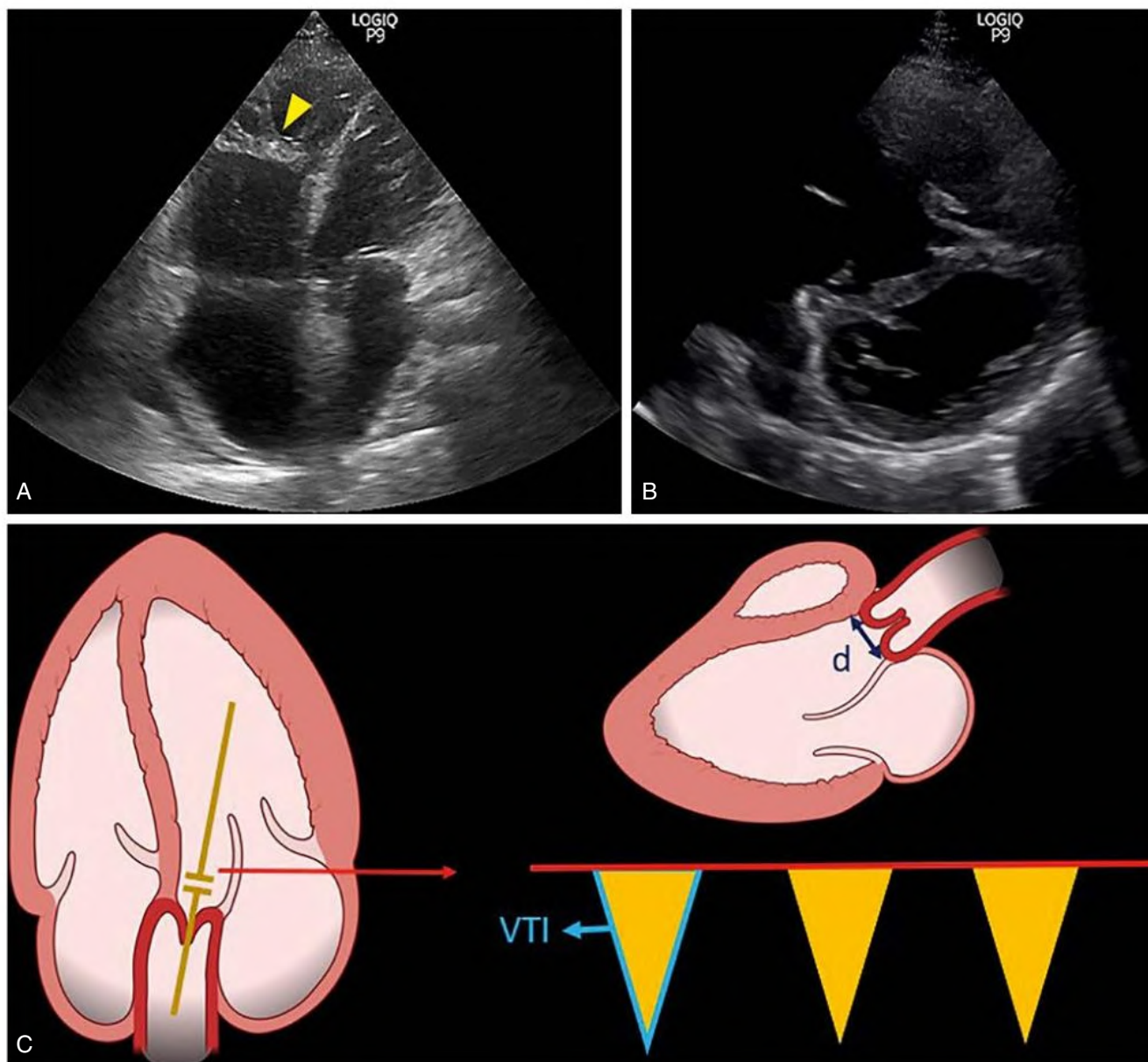


Fig. 5.11 Apical four-chamber (A) and parasternal short axis (B) views obtained from a patient with severe pulmonary hypertension. Note severe enlargement of the right ventricle (larger than left ventricle in apical view) and interventricular septal flattening with D-sign on short axis. Arrowhead points to hypertrophied moderator band in the right ventricle. (C) Components of stroke volume assessment. Velocity time integral (VTI) is obtained by tracing the pulsed wave Doppler envelope of the left ventricular outflow tract (LVOT) in apical five-chamber view. Cross sectional area of the LVOT is obtained by measuring its diameter (d) at the base of the aortic valve in midsystole and using the formula for area of a circle, πr^2 (where r = radius, or diameter divided by 2). (Created with [Biorender.com](#).)

leg raise is a simple bedside method of predicting fluid responsiveness. It is performed by tilting the patient from a 45-degree semirecumbent head-up position to a 45-degree leg-up position, which transfers up to 300 to 400 mL of blood into the central circulation and acting as an autogenous fluid bolus. Stroke volume is measured at the LVOT before and 1 minute after the leg raise; an increase by 10% suggests fluid responsiveness.¹⁴ Stroke volume is the product of the cross-sectional area (CSA) of the LVOT and the velocity time integral (VTI). LVOT CSA is derived from the LVOT diameter in PLAX view using the formula πr^2 , and LVOT VTI is obtained by tracing the envelope of the Doppler spectrum of systolic flow in the apical five-chamber view (Fig. 5.11C). As the LVOT diameter is constant for a given person, VTI

(normal: approximately 18–22 cm) is generally used as a surrogate for cardiac output changes, assuming the heart rate is relatively constant. This parameter is also used to differentiate between true hypovolemia (VTI will be low) and high cardiac output states (VTI will be high); both conditions are often associated with a hyperdynamic left ventricle and small IVC.

ULTRASOUND OF THE LUNG AND PLEURA

Because air transmits ultrasound poorly, the normally aerated lung parenchyma prevents the visualization of anatomic structures beyond the interface between the parietal and the visceral pleura. Nonetheless,

ultrasound can provide important information related to the pleural and subpleural space. A key feature of lung ultrasound is that it relies on the interpretation of artifacts, which are nonanatomic images generated by the interaction between ultrasound and tissues.

Technique

Lung and pleural ultrasound can be performed using most entry-level POCUS devices. Linear (vascular probe), phased array (cardiac), and curvilinear (abdominal) transducers may be used. However, the preferred transducer will depend on the type of pathology, and multiple types of transducers may be required. The linear transducer has improved resolution of superficial structures, which makes it better for assessing the chest wall, pleura, and superficial subpleural space. Phased array and curvilinear transducers can image structures at a higher depth, which make them superior for imaging lung consolidation and pleural effusion. All transducers are able to visualize artifacts arising from below the pleural line. However, phased array and curvilinear transducers may be superior to linear transducers for this indication.¹⁵ Ultrasound manufacturers usually include a preconfigured setting for lung ultrasound.

To adequately visualize the structures and to reduce artifacts, it is important to position the probe at a perpendicular (90-degree) angle to the chest. This usually produces normal A-line artifacts, as shown in Fig. 5.12A–B. Using a cephalocaudal orientation, the rib shadows help with the identification of intercostal spaces. Because lung ultrasound only provides information at the site where the transducer is located, multiple sites of assessment on the thorax are always required: protocols include assessment of 28 zones (16 right zones and 12 left zones),¹⁶ eight zones (four on each side),¹⁷ or six zones (three on each side).¹⁸ Multiple factors related to the clinical context must be considered when deciding the number of zones to assess. For example, a six-zone protocol might be sufficient for the rapid assessment of an acutely dyspneic patient (Fig. 5.13).¹⁸ On the other hand, evaluation of lung congestion in asymptomatic or minimally symptomatic patients may require more than six zones.¹⁶ Although the evidence is limited, the use of simpler protocols (eight zones) may be comparable to the more time-consuming 28-zone approach.¹⁷

Interpretation

After adequate visualization of an intercostal space (the area between two rib shadows), the first step is to examine the hyperechoic pleural line just deep to intercostal muscles. This line oscillates or shimmers with respiration, which represents pleural sliding. The presence of sliding confirms that air is not present between the parietal and visceral pleurae (Video 5.3). Movement transmitted from the heart, commonly referred to as a *lung pulse*, also serves the same purpose. This can rule out the presence of a pneumothorax or subcutaneous emphysema at the site of assessment. It is important to first exclude these diagnoses because air will preclude the assessment of deeper structures and produce a pseudonormal A-line pattern that could mislead. Lung sliding and lung pulse can be better visualized by decreasing the depth of the image and by using a linear transducer.

After confirmation of the pleural sliding, depth should be increased up to 10 to 12 cm below the pleural line to examine the subpleural space and optimize the visualization of artifacts, which are illustrated in Fig. 5.12A–D. The normal A-line pattern represents linear artifacts that are parallel and equidistant from the pleural line. These are created by ultrasound reverberation from the pleural interface. B-lines represent linear artifacts that arise from the pleural line, extend to the end of the image without fading, and move synchronously with lung sliding. They tend to completely or partially erase A-line in their path (Video 5.4). B-lines can become *confluent* or *coalescent*, completely erasing

the A-lines and giving a *white lung* appearance in cases where there is severe loss of lung aeration. B-lines are the hallmark of increased density of lung parenchyma in the first millimeters below the pleural interface with partial aeration and generally represent increased extravascular lung water. However, they can be seen in multiple pathologies as summarized in Table 5.1. One or two isolated B-line artifacts per rib interspace may not indicate pathology, particularly in the dependent lung zones. The visualization of three or more B-lines in a rib interspace (a *B-line region*) is often significant, and more than two positive B-line regions bilaterally constitute *interstitial syndrome*. The distribution of B-lines, associated ultrasound findings, and clinical context may help differentiate between possible etiologies.

Consolidated or atelectatic lung parenchyma will transmit ultrasound because it is devoid of air and will produce variable appearance on ultrasound, from a *hypoechoic* region to a tissue or liver-like appearance (*hepatization*) (Fig. 5.12E–F). It is important to note that central consolidations that do not reach the pleura are not detected by ultrasound because air is present between the pathology and the transducer. Because of this, lung ultrasound has variable sensitivity for the diagnosis of pneumonia.

Pleural effusion appears as an *anechoic* space above the diaphragm, usually surrounding the atelectatic lung. Effusions can be identified, characterized, and quantified using POCUS (Fig. 5.12G–H). Detection of small pleural effusions require the phased array or curvilinear transducer to be positioned in the midaxillary line to obtain a coronal view of the interface between the lung, the diaphragm, and the liver/spleen. Seeing the dorsal vertebral column in this position usually implies the presence of fluid permitting the transmission of sound. This is called the *spine sign* and is the most sensitive sign to detect a small pleural effusion. Unlike uniformly anechoic simple effusions, complex effusions often contain swirling internal echoes (*plankton sign*) and may demonstrate fibrin stranding or distinct septations. Although most complex effusions tend to be of infectious etiology, a hemothorax with partially clotted blood can have a similar appearance and should be interpreted in the appropriate clinical context.

Clinical Applications

Lung and pleural ultrasound is a valuable adjunct to physical examination in the management of patients with undifferentiated dyspnea and those with difficult-to-determine fluid status. Both of these clinical scenarios are frequently encountered in the care of patients with kidney disease, including those on kidney replacement therapy.

In patients with dyspnea, POCUS is highly sensitive to detect pneumothorax,¹⁹ pleural effusions,²⁰ and diffuse interstitial syndromes such as acute pulmonary edema,²¹ acute respiratory distress syndrome (ARDS), and diffuse viral pneumonia²² (see Table 5.1). Subpleural consolidations, an irregular pleural line, or a patchy involvement with spared areas all suggest diffuse pneumonia or ARDS, whereas a homogeneous distribution of B-lines without pleural anomalies is more compatible with pulmonary edema. B-lines observed in a single location on the thorax may indicate localized pneumonia, partial atelectasis, or pulmonary embolism. However, the sensitivity of lung ultrasound is moderate for bacterial pneumonia²³ and low for pulmonary embolism.²⁴ These pathologies may present with dyspnea in the presence of primarily an A-line pattern with lung sliding. An A-line pattern will also be seen in acute exacerbations of asthma or chronic obstructive pulmonary disease. A diagnostic algorithm based on the bedside lung ultrasound in emergency (BLUE) protocol¹⁸ used in the intensive care and emergency medicine settings is presented in Fig. 5.13.

Even in the absence of dyspnea, pulmonary congestion may be present, and its detection provides important information to guide fluid management decisions. For example, B-lines are detected in a significant

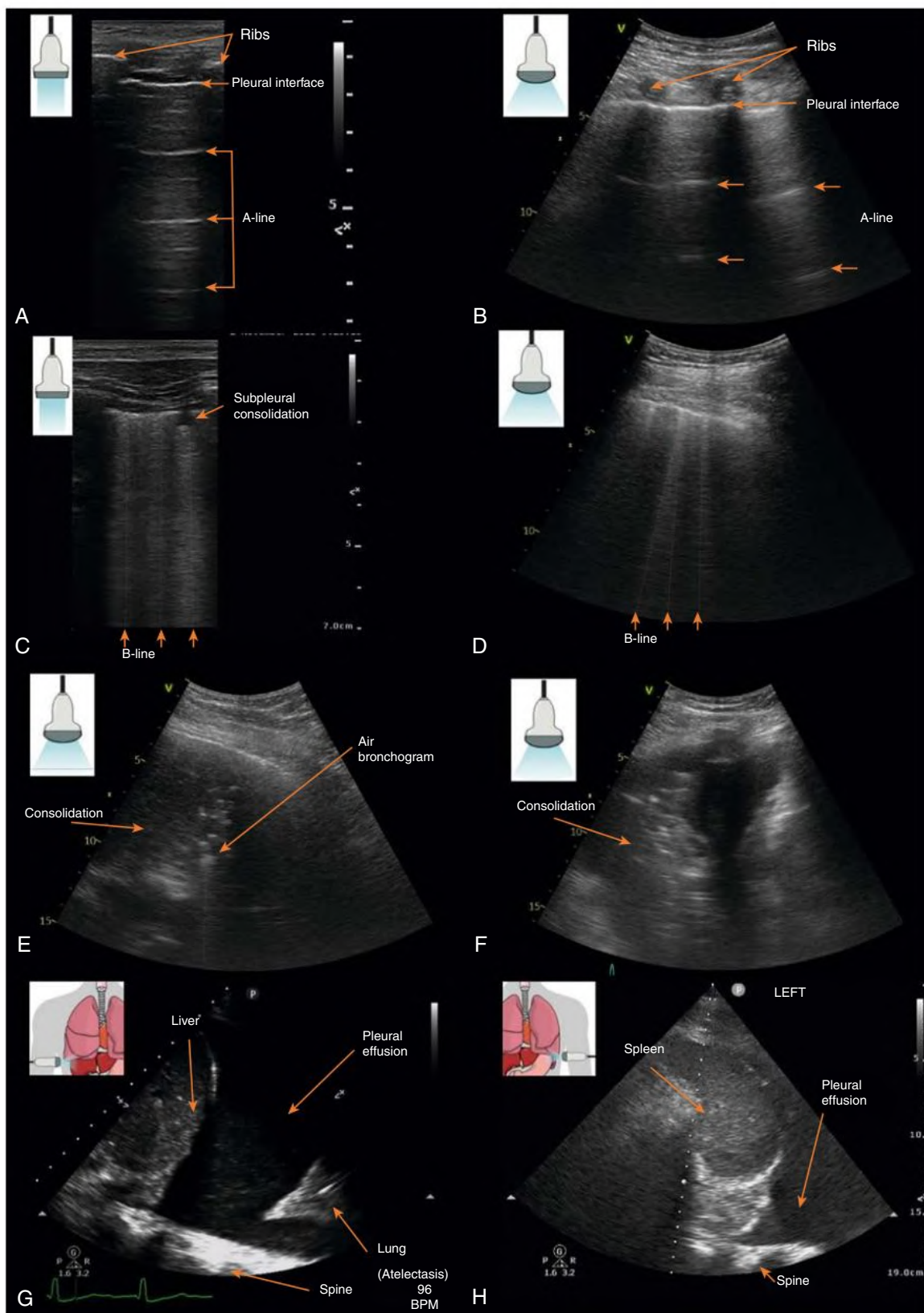


Fig. 5.12 Examples of Lung and Pleural Ultrasound. Normal ultrasound (A-line pattern) using a linear transducer (A) and a curvilinear transducer (B). B-lines shown using a linear transducer (C) and a curvilinear transducer (D). Note that B-lines move with respiration (see Video 5.4). Examples of lung consolidation (also referred to as the C-pattern) showing a heterogeneous tissue-like appearance (E–F). Pleural effusion of the right lower thorax region (G) and to the left lower thorax region (H).

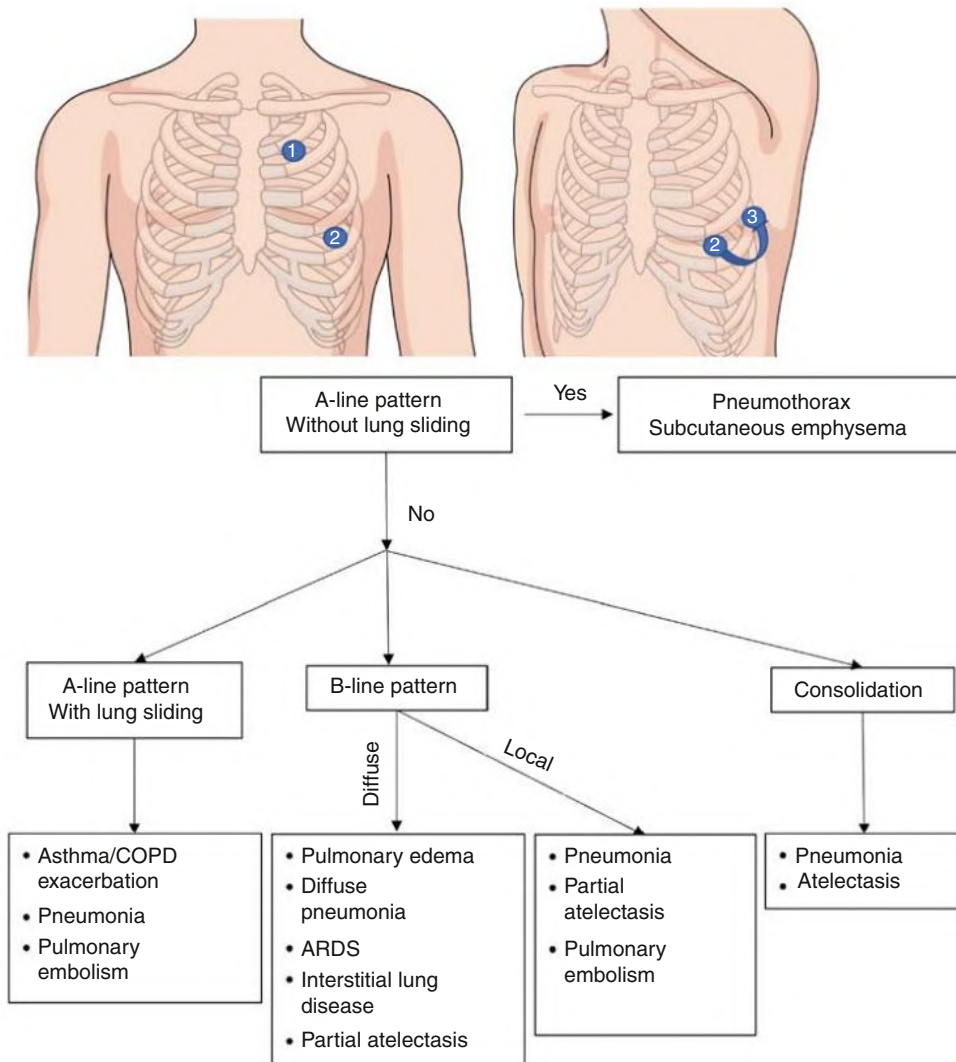


Fig. 5.13 Proposed Approach to Acute Dyspnea Using Lung Ultrasound Based on the Bedside Lung Ultrasound in Emergency (BLUE) Protocol.⁴ Three zones on each side are assessed: the upper point (1), the lower point (2), and the posterolateral point (3). The original BLUE protocol also includes assessment of lower extremity veins to detect deep venous thrombosis suggestive of pulmonary embolism. ARDS, Acute respiratory distress syndrome; COPD, chronic obstructive pulmonary disease.

proportion of asymptomatic patients with end-stage kidney disease²⁵ and chronic heart failure, and B-lines appear to be more sensitive than chest auscultation to identify pulmonary congestion.²⁶ In hemodialysis patients, the B-line count is reduced in proportion to the amount of intradialytic ultrafiltration²⁷ and seems to correlate with mortality,^{16,17,28} although it is unclear whether modifying fluid removal prescription to target resolution of B-lines improves outcomes.²⁹ A recent small trial showed that target weight prescription guided by lung ultrasound improved blood pressure without increasing the risk of intradialytic hypotension³⁰ and improved LV filling pressures by echocardiography.³¹ Interestingly, two single-center trials performed in ambulatory chronic heart failure patients showed that the addition of lung ultrasound to the physical examination may reduce the risk of hospitalization.^{32,33}

ARTERIAL AND VENOUS DOPPLER OF THE KIDNEY AND OTHER ORGANS

Doppler ultrasonography enables assessment of blood flow velocity and patterns during the cardiac cycle at various locations in the circulatory

system. Arterial Doppler waveforms are influenced by upstream and downstream factors including pulse pressure, the presence of stenosis, blood flow, and vascular resistance of the downstream vascular bed. Venous Doppler waveforms are primarily influenced by the pattern of venous return during the cardiac cycle and systemic venous compliance. The next sections provide a brief overview of this topic; a more detailed account can be found elsewhere.^{34,35}

Arterial Doppler

A comprehensive ultrasound examination to evaluate suspected renal artery stenosis involves visualizing the renal arteries throughout their course from the aorta to the level of the segmental and interlobar arteries. In addition, accurate flow velocities need to be measured at multiple sites, which depends on an appropriate angle of insonation and requires a higher operator skill level. Therefore it is better performed by a trained sonographer, rather than a physician, using POCUS to answer focused questions at the bedside.

On the other hand, renal resistive index (RI), which is an indirect indicator of renal artery stenosis, is a ratio and therefore is less

TABLE 5.1 Differential Diagnosis of B-Line Artifacts on Lung Ultrasound

Etiology of B-Lines	Associated Findings/Comments
Diffuse Pattern	
Pulmonary edema	Higher severity in dependent lung regions
Viral pneumonia	Nonhomogeneous distribution (some areas are “spared”) with irregular pleural line and subpleural consolidations ^a
ARDS	Nonhomogeneous distribution with irregular pleural line and subpleural consolidations
Interstitial lung disease	Nonhomogeneous distribution irregular pleural line and subpleural consolidations
Local Pattern	
Localized pneumonia	Sensitivity is imperfect because the pneumonia must extend to the pleural interface before ultrasound can detect it Pleural sliding often decreased/absent
Partial atelectasis	Pleural sliding often decreased/absent
Pulmonary embolism	Low sensitivity

Lung ultrasound findings must be interpreted in the appropriate clinical context and in conjunction with physical examination/other sonographic findings. For example, history might indicate preexisting interstitial lung disease or suggest an infectious etiology in a patient presenting with fever, etc.

^aPleural line irregularities and subpleural consolidations are better appreciated using a linear transducer.

dependent on the angle of insonation. RI is calculated from intrarenal arterial (typically interlobar) Doppler tracing using the following formula:

$$(\text{Peak systolic velocity} - \text{End diastolic velocity}) / \text{Peak systolic velocity}$$

The normal range is approximately 0.50 to 0.70. Despite its name, the renal vascular resistance is only one factor affecting the RI, and the interpretation must take into consideration the influence of systemic hemodynamics. Hemodynamically significant renal artery stenosis is often associated with a lower RI in the affected kidney. A side-to-side difference of greater than 0.05 is highly suggestive of unilateral renal artery stenosis and should prompt further investigations to confirm the diagnosis. RI may also provide prognostic information in this context; the presence of a high RI (>0.80) is associated with a lower chance of response to revascularization.³⁶ Similarly, in kidney transplant

recipients, an RI of 0.80 or greater has been shown to be associated with increased incidence of mortality.³⁷ Elevated renal RI is also found in advanced chronic kidney disease and acute kidney injury and does not inform on the cause of kidney injury. Moreover, it is uncertain whether an elevated RI can help to predict the likelihood of recovery from acute kidney injury.^{38–40}

Venous Doppler

The normal Doppler tracing of portal and intrarenal veins (usually interlobar) is relatively continuous, whereas hepatic vein Doppler is triphasic with a dominant systolic component. Portal waveform is above the baseline, as the flow is toward the transducer, whereas in renal veins it is below the baseline, as the flow is away from the transducer while acquiring the images. Similarly, the systolic and diastolic waves of the hepatic vein are below the baseline. Intrarenal Doppler can be technically challenging because the vessels are small, and the pulsed wave Doppler sample volume gets displaced with respiratory movement of the kidney.

When the RA pressure increases, venous return during systole is reduced (because the RA is already distended), and the venous circulation become distended and thus less compliant. This results in the alteration of the associated venous waveforms. First, the usually dominant systolic wave on hepatic vein Doppler is progressively reduced and even reversed in severe cases. Next, portal and intrarenal vein waveforms become increasingly pulsatile, ultimately resulting in flow reversal during systole when RA pressure is severely elevated. In other words, in cases of severe congestion, venous return occurs only during diastole (Fig. 5.14). Interestingly, these changes can be reversed with measures aimed at reducing RA pressure, such as diuretic or ultrafiltration therapy, and can be monitored in real time.

These venous Doppler flow alterations have been associated with adverse outcomes in congestive heart failure,^{41,42} after cardiac surgery,^{43,44} and in general critically ill patients.⁴⁵ A venous excess ultrasound (VExUS) grading system combining the presence of these abnormal venous Doppler signals may have some advantages in certain clinical settings (e.g., significant tricuspid regurgitation, ventricular pacing, portal hypertension) and correlates with the risk of acute kidney injury in a cohort of cardiac surgery patients.⁴⁶ However, it is unknown whether using these sonographic markers to guide management will improve outcomes.

Not all dyspnea is due to extravascular lung water, nor is all hypotension due to hypovolemia. Therefore the use of multiorgan POCUS in conjunction with conventional clinical assessment aids in diagnosis. Table 5.2 summarizes typical POCUS findings in major types of shock (hypovolemic, cardiogenic, obstructive, and distributive). The same applies to undifferentiated hypotension without overt shock. However, it is not uncommon to encounter overlapping etiologies in clinical practice; therefore these findings should not be considered absolute.

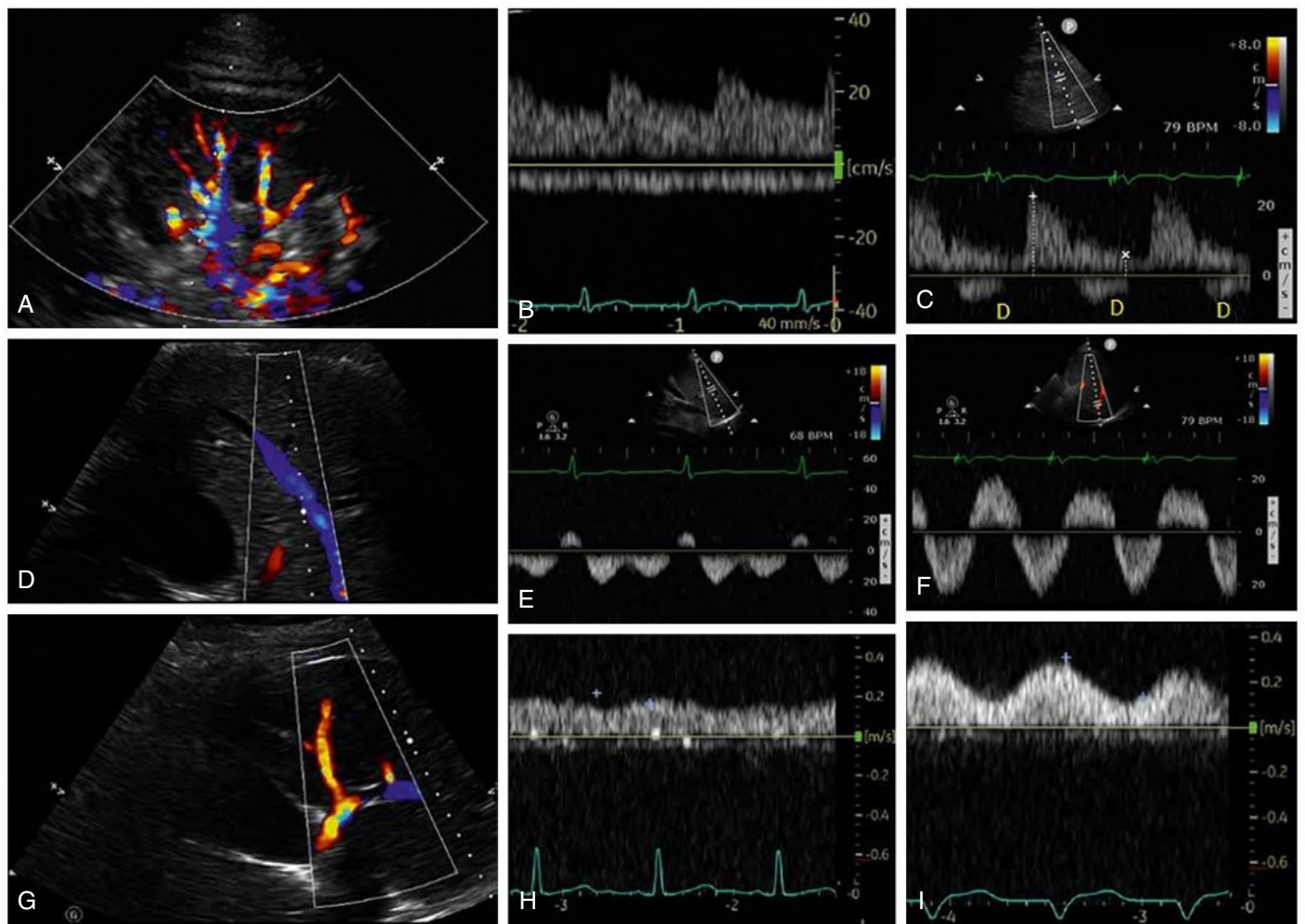


Fig. 5.14 Examples of Normal and Abnormal Venous Doppler Waveforms. (A) Color Doppler image showing the site of intrarenal Doppler assessment. (B) Normal intrarenal Doppler showing a normal arterial resistive index (above the baseline) and a continuous venous signal during the cardiac cycle (below the baseline). (C) Abnormal intrarenal Doppler showing both a high resistive index and discontinued venous signal only present in diastole (indicated by letter D). (D) Color Doppler image showing the right hepatic vein. (E) Normal hepatic vein Doppler with dominant systolic component. (F) Abnormal hepatic vein flow with reverse systolic component and only D-wave below the baseline. (G) Color Doppler image showing the main portal vein. (H) Normal portal vein Doppler showing minimal (<30%) variation in velocities during the cardiac cycle. (I) Abnormal portal vein Doppler with pulsatility ($\geq 50\%$ variation) during the cardiac cycle.

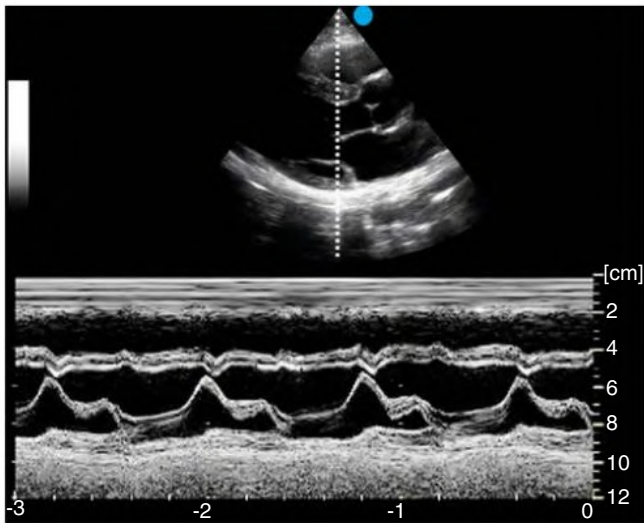
TABLE 5.2 Characteristic Point-of-Care Ultrasound Findings in Major Types of Shock

Type of Shock	Hypovolemic	Cardiogenic	Obstructive	Distributive
Hemodynamic mechanism	Volume depletion	Pump failure	Obstruction of pump function (e.g., pulmonary embolism, pericardial effusion)	Vasodilation leading to improper distribution of cardiac output resulting in impaired organ perfusion
Lung ultrasound	A-lines	Diffuse B-lines Pleural effusions	A-lines Focal B-lines may be seen in lung infarction	A-lines Pleural line irregularities with diffuse B-lines in ARDS Consolidation in pneumonia
FoCUS including IVC	Hyperdynamic LV (increased <i>ejection</i>) Decreased stroke volume (<i>exit</i>) Small, collapsible IVC (<i>entry</i>)	Decreased LV <i>ejection</i> Decreased cardiac output (<i>exit</i>) Plethoric IVC (<i>entry</i>) Chamber enlargement in some cases (altered <i>equality</i>)	Pericardial <i>effusion</i> Dilated RV in pulmonary embolism (altered <i>equality</i>) Decreased cardiac output (<i>exit</i>) Plethoric IVC (<i>entry</i>)	Hyperdynamic LV (increased <i>ejection</i>) with increased cardiac output (<i>exit</i>) Decreased LV function in septic cardiomyopathy IVC variable (<i>entry</i>)
Venous Doppler	Hepatic: S > D Portal: continuous Intrarenal: continuous	Hepatic: S < D, possible S reversal Portal: pulsatile, possible systolic flow reversal Intrarenal: Pulsatile, possible systolic flow reversal	Variable depending on the etiology	Similar to hypovolemic if volume depleted Similar to cardiogenic if excessively volume resuscitated or cardiomyopathy
Others	Intraperitoneal free fluid in case of hemorrhage	Ascites Bowel wall edema	Deep vein thrombosis in case of pulmonary embolism	Source of sepsis (e.g., cholecystitis, pyonephrosis)

ARDS, Acute respiratory stress syndrome; FoCUS, focused cardiac ultrasound; IVC, inferior vena cava; LV, left ventricle; RV, right ventricle; S, systolic wave; D, diastolic wave.

SELF-ASSESSMENT QUESTIONS

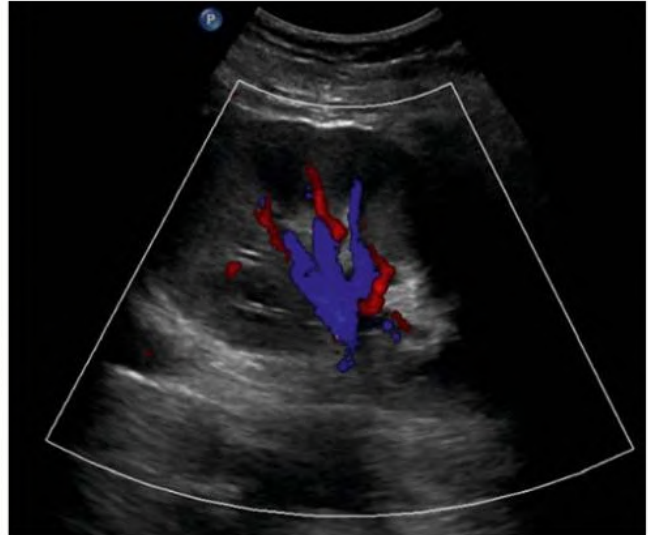
1. Which ultrasound mode is used to obtain the following image?



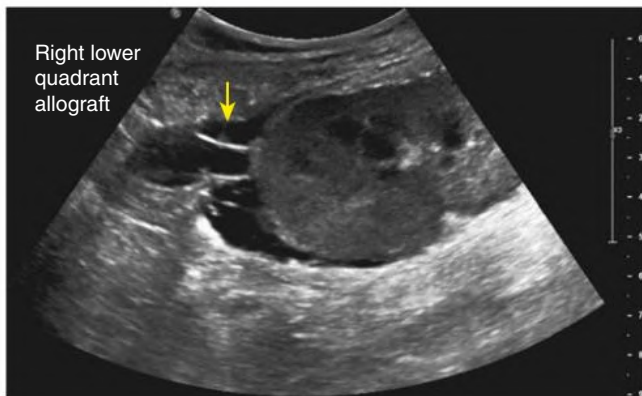
- A. Color Doppler
- B. M-mode
- C. Pulsed wave Doppler
- D. Continuous wave Doppler

2. Below are the grayscale and color Doppler images of the left kidney obtained from a patient with AKI. What is the sonographic finding?

- A. Hydronephrosis
- B. Parapelvic cyst
- C. Prominent blood vessels
- D. Hematoma



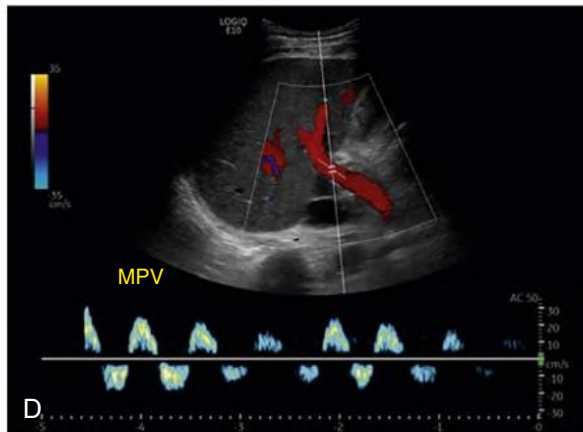
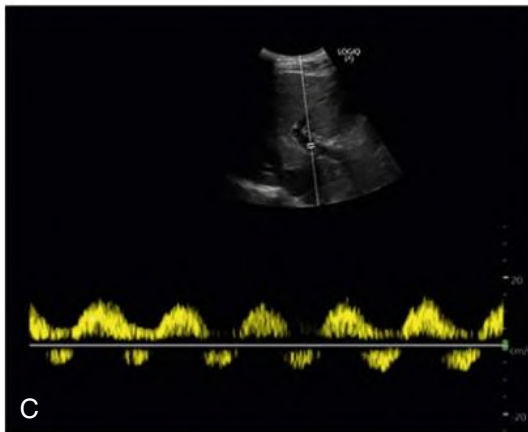
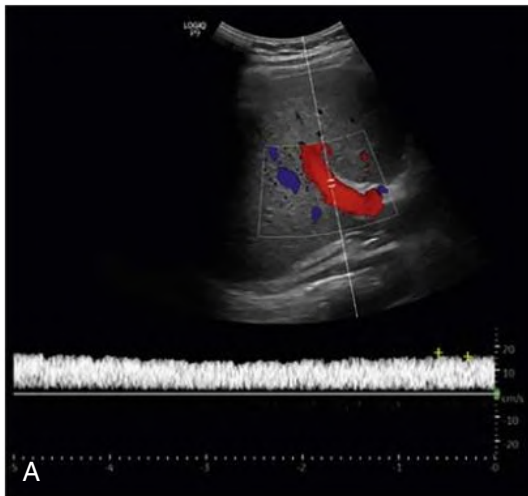
3. You are evaluating a renal transplant recipient in early postoperative period. The allograft demonstrates an abnormal appearing area at the upper pole (arrow). What is the likely diagnosis?



- A. Hematoma
B. Urinoma
C. Hydronephrosis
D. Pelvic ascites
4. B-lines on lung ultrasound can be seen in which of the following conditions?
- A. Pulmonary edema
B. Pulmonary fibrosis
C. Pneumonia
D. All the above

5. Which of the following images represents normal portal vein Doppler waveform?

- A. A
B. B
C. C
D. D



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